

AUSTIN 33 NW, TX, USA

WMO: 747440

Lat: 30.6222N

Lon: 98.0846W

Elev: 415

StdP: 96.44

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 23907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-1.6	-12.1	1.4	1.7	-9.9	1.7	3.6	6.5	7.2	5.9	7.9	2.7	N/A	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	36.9	21.7	35.8	21.9	34.9	22.0	24.4	30.4	24.0	30.0	23.7	29.6	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	18.8	25.4	22.8	18.4	25.3	22.3	17.9	25.3	75.7	30.7	74.3	29.9	72.9	29.6	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.6	4.9	4.3	-7.5	39.2	2.3	1.6	-9.1	40.3	-10.4	41.2	-11.7	42.1	-13.3	43.3
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	9.8	12.2	16.0	19.7	23.3	27.3	28.5	29.0	25.8	20.7	15.0	10.7
	DBStd	7.92	5.54	6.16	4.95	3.74	3.27	1.97	1.62	1.95	2.84	4.38	5.11	5.38
	HDD10.0	204	72	44	12	0	0	0	0	0	0	2	16	58
	HDD18.3	974	266	182	101	29	5	0	0	0	1	29	120	242
	CDD10.0	3809	66	106	199	292	412	518	574	588	475	332	167	79
	CDD18.3	1536	2	10	29	71	159	268	316	330	226	101	20	4
	CDH23.3	15304	16	56	177	484	1253	2838	3649	3977	2058	698	84	13
	CDH26.7	7183	1	10	27	119	446	1363	1926	2172	918	193	9	0
Wind	WSAvg	2.1	2.1	2.3	2.6	2.6	2.3	2.0	1.8	1.7	1.6	1.9	2.0	2.1
Precipitation	PrecAvg	840	52	54	71	72	79	92	56	50	84	84	72	56
	PrecMax	1203	134	160	219	183	177	287	234	155	217	290	272	247
	PrecMin	543	1	3	15	7	16	4	0	0	1	0	1	2
	PrecStd	202	40	42	44	45	47	69	55	43	53	72	70	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	28.3	29.3	32.2	34.2	37.6	38.0	38.5	36.4	32.6	28.1	25.2
		MCWB	14.4	14.5	17.1	19.2	21.6	21.7	21.3	21.7	20.0	20.9	18.9	17.0
	2%	DB	22.7	24.9	27.0	29.7	32.4	35.4	36.6	37.1	34.6	30.4	25.7	22.7
		MCWB	14.0	14.5	16.7	19.2	21.3	22.2	21.6	22.2	21.0	21.4	18.2	16.2
	5%	DB	20.2	22.6	25.3	27.9	30.8	34.1	35.4	36.2	33.1	28.9	23.9	20.7
		MCWB	13.9	15.9	16.6	18.6	21.0	22.1	21.8	22.3	21.3	20.2	17.6	15.0
	10%	DB	18.0	20.6	23.6	26.3	29.3	32.9	34.4	35.1	31.7	27.4	22.1	19.0
		MCWB	12.5	15.6	16.0	18.1	20.8	22.3	22.3	22.3	21.2	18.9	17.4	15.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.7	20.1	22.8	24.1	24.9	24.7	24.7	24.5	23.4	21.7	19.5
		MCDB	20.5	22.5	24.5	27.4	28.3	30.6	30.4	31.9	29.5	27.9	25.1	22.4
	2%	WB	17.4	18.8	19.3	21.6	23.3	24.2	24.3	24.1	23.9	22.6	20.7	18.5
		MCDB	20.6	21.1	23.1	26.0	27.8	30.2	30.2	31.3	28.9	26.8	23.5	20.9
	5%	WB	15.5	17.9	18.7	20.8	22.7	23.8	23.9	23.8	23.5	22.0	19.6	17.5
		MCDB	18.4	20.0	22.3	25.0	27.3	29.7	29.8	30.8	28.2	25.6	22.5	19.7
	10%	WB	13.4	16.8	18.0	20.1	22.1	23.4	23.5	23.4	23.1	21.4	18.6	15.6
		MCDB	17.4	19.4	21.6	24.2	26.7	29.1	29.3	30.3	27.5	25.3	21.1	17.9
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.8	11.2	10.9	10.1	10.6	11.4	11.7	10.3	10.5	10.0	9.8
		MCDBR	12.9	12.1	12.3	12.4	12.0	12.1	12.9	13.0	12.1	11.0	10.6	11.5
		MCWBR	7.4	6.7	5.8	5.6	4.3	2.7	2.6	2.6	3.4	4.1	5.6	7.0
	5% WB	MCDBR	10.8	8.8	10.0	9.9	9.3	9.9	10.4	11.5	9.4	9.5	8.9	9.8
	MCWBR	8.1	6.9	6.0	5.7	4.0	3.0	2.9	2.6	2.9	4.7	6.0	8.2	
Clear-Sky Solar Irradiance	taub	0.310	0.319	0.345	0.384	0.419	0.410	0.422	0.413	0.389	0.361	0.339	0.311	
	taud	2.533	2.514	2.452	2.363	2.315	2.384	2.392	2.396	2.413	2.478	2.478	2.538	
	Ebn at Noon	926	947	940	909	874	876	864	870	883	888	882	907	
	Edh at Noon	86	96	109	124	131	122	120	118	112	98	90	82	
All-Sky Solar Radiation	RadAvg	3.07	3.60	4.61	5.53	5.83	6.69	6.72	6.33	5.21	4.25	3.23	2.72	
	RadStd	0.33	0.59	0.42	0.50	0.57	0.48	0.54	0.35	0.48	0.62	0.40	0.36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page